BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT

(An Autonomous Institution Affiliated to Visvesvaraya Technological University, Belagavi)

Department of Master of Computer Applications (MCA) • Accredited by NBA

PROJECT WORK PRESENTATION

Digital Platform for Direct Marketing of Local Farm Produce

An Integrated Dual-Capability Digital Agritech Ecosystem: FarmFetch Marketplace + KrishiMitra AI Assistant

🌾 **Direct Farm-to-Consumer Market**

🤖 **WhatsApp KrishiMitra AI Agronomist**

🔐 **Dual-OTP Fraud-Proof Logistics**

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Introduction: FarmFetch + KrishiMitra AI

An integrated dual-capability digital agricultural ecosystem bridging local producers and end consumers.

CORE ARCHITECTURAL CONCEPT

A unified, dual-capability digital agricultural platform designed to bridge the structural gap between local producers and end consumers by coupling a hyper-local direct trade marketplace with a voice-enabled, multimodal AI agronomy assistant.

1. Farm-to-Consumer Marketplace

- **Hyper-Local Farm Discovery:** Allows consumers to detect and explore nearby peri-urban farms based on live GPS location and radius filtering.
- **Direct Farm-Gate Procurement:** Empowers farmers to set transparent pricing and sell directly to consumers without intermediary Mandi cuts.
- **Real-Time Order Tracking:** Integrated Leaflet.js interactive maps and Flask-SocketIO track orders seamlessly from farm harvest to doorstep.
- **Atomic Stock Protection:** Prevents inventory overselling during peak checkout bursts via database-level transactional locks.
- **Double OTP Validation:** Cryptographically secure 4-digit pickup and delivery handshakes protecting both farmer and customer funds.

2. KrishiMitra AI Assistant

- **Conversational WhatsApp Gateway:** Eliminates application install friction; accessible over Meta WhatsApp Cloud API for rural usability.
- **Instant Leaf Disease Diagnosis:** Computer vision pathology engine detects leaf blights, powdery mildew, and pests from uploaded plant photographs.
- **Multilingual Voice Assistance:** Powered by OpenAI Whisper STT and Microsoft Edge-TTS, supporting spoken regional dialect queries.
- **Contextual Agronomic Advice:** Ultra-fast Groq LPU inference (Qwen model) delivers tailored organic remedies, dosage, and preventive schedules.
- **Schemaless Query Logging:** MongoDB document storage retains multimodal conversation threads and diagnosis logs for longitudinal crop analytics.

The Problem: Intermediaries & Lack of Technical Aid

Analyzing the economic exploitation, diagnostic delays, and digital silos stifling agricultural profitability.

1. THE INTERMEDIARY TRAP

Economic Exploitation by Middlemen

- **Mandi Monopoly:** Traditional agriculture relies on layers of middlemen, commission agents, and Mandi traders who control price discovery.
- **Erosion of Farmer Incomes:** Farmers routinely receive less than 25-30% of the ultimate consumer price, barely covering seed and fertilizer input costs.
- **Inflated Consumer Prices:** Multiple handling fees, commissions, and logistics markups drastically inflate retail grocery bills.
- **Opaque Transactions:** Arbitrary grading deductions, weight manipulation, and prolonged credit cycles leave farmers vulnerable.

2. THE KNOWLEDGE GAP

Severe Lack of Timely Agronomic Aid

- **Agronomist Scarcity:** Rural areas suffer from severe shortages of agricultural extension officers, leaving smallholders without scientific advice.
- **Delayed Pathology Diagnosis:** Crop infections and fungal blights escalate exponentially within 48-72 hours; slow manual diagnosis leads to 30-50% crop destruction.
- **Indiscriminate Chemical Usage:** Farmers rely on guesswork or pesticide retail advice, causing toxic over-spraying, soil degradation, and wasted capital.
- **Literacy & Tech Barriers:** Traditional mobile apps demand English typing, complex navigation, and high digital fluency unsuited for rural users.

3. SYSTEM FRAGMENTATION

Isolated Software Silos

- **Disconnected Platforms:** Existing software is strictly bifurcated—handling either macro wholesale trading or standalone computer vision research, but never both.
- **No Unified Pathway:** Even if a farmer identifies and cures a crop infection using a standalone model, they lack an immediate digital sales channel to commercialize produce.
- **High App Friction:** Farmers abandon complex multi-screen mobile apps that demand separate logins, distinct wallets, and redundant data entry.
- **Zero Delivery Coordination:** Wholesale platforms lack hyper-local last-mile dispatch, leaving direct deliveries to chaotic ad-hoc phone calls.

Literature Review & Competitive Differentiation

Comparative benchmarking highlighting why FarmFetch + KrishiMitra AI overcomes standard market deficiencies.

⚠️ EXISTING AGRICULTURAL SYSTEMS & GAPS

- **National Agriculture Market (e-NAM):**
  - Focuses strictly on macro-level inter-Mandi wholesale auctions and bulk trade.
  - Completely ignores hyper-local consumer delivery and peri-urban farm discovery.
  - High regulatory onboarding overhead; inaccessible to smallholder farm-gate sales.
- **Standalone Deep Learning CNN Models:**
  - Academic research models (ResNet, MobileNet) accurately classify leaf disease datasets (e.g. PlantVillage).
  - Remain trapped in research sandboxes without integrated commercial marketplaces.
  - Provides no channel for farmers to sell healthy yields or recover treatment expenses.
- **Commercial Grocery Platforms (Blinkit / BigBasket):**
  - Centralized dark-store model sourcing from aggregate vendors; farmers remain price-takers.
  - No direct farmer-consumer interaction, zero producer branding, and heavy retail commission cuts.

✅ WHAT MAKES OUR PROJECT UNIQUE

- **Unified Dual-Capability Ecosystem:**
  - Seamlessly integrates an automated transactional marketplace with advanced conversational AI under one roof.
  - Farmers seamlessly transition from crop pathology diagnosis to harvesting and instant direct listing.
- **Double OTP Fraud Prevention Architecture:**
  - Enforces strict accountability across transit via a dual-handshake protocol.
  - Pickup OTP (Farmer ➔ Driver): Guarantees driver collected correct produce at the farm.
  - Delivery OTP (Customer ➔ Driver): Confirms doorstep fulfillment before releasing funds.
- **Automated Split Wallet Settlements:**
  - Programmatic, real-time escrow split on verified order delivery.
  - Example Settlement Rule: 92% credited instantly to Farmer's wallet, 85% of delivery fee to Driver, and 8% retained for platform operations—zero delayed credit cycles.

ARCHITECTURE & ACCESS CONTROL

System Architecture & Role-Based Access Control (RBAC)

Granular access control, microservice boundaries, and role responsibilities across 4 platform actors.

FARMER (PRODUCER)

- **Digital Catalogue & Inventory:** Publishes produce listings with live pricing, harvest timestamps, and stock thresholds.
- **Order Management:** Receives real-time browser audio pings via WebSockets on order placement.
- **Pickup Verification:** Validates driver identity using 4-digit Pickup OTP before releasing harvest.
- **Financial Realization:** Monitors digital wallet balance with instant post-delivery credits.

CUSTOMER (CONSUMER)

- **Location-Aware Discovery:** Discovers local farms based on geographic radius and current browser coordinates.
- **Cart & Seamless Checkout:** Builds fresh produce carts with atomic quantity reservation.
- **Live Delivery Tracking:** Tracks driver dispatch in real time across dynamic Leaflet.js map routes.
- **Delivery OTP Handshake:** Supplies 4-digit Delivery OTP upon inspecting produce freshness at doorstep.

DELIVERY DRIVER (LOGISTICS)

- **Local Order Dispatch Feed:** Reviews available deliveries within localized geographical proximity.
- **Interactive GPS Navigation:** Follows optimized waypoint routes from farm-gate to customer doorstep.
- **Dual-Step Handshake:** Submits Pickup OTP to farmer and validates Delivery OTP from customer.
- **Transparent Earnings:** Earns immediate 85% delivery commission credited directly into driver wallet.

PLATFORM ADMINISTRATOR

- **Governance & Verification:** Audits farm KYC credentials, verifies business compliance, and approves accounts.
- **System-Wide Telemetry:** Monitors transaction volumes, order fulfillment rates, and delivery latencies.
- **Dispute Arbitration:** Inspects transaction audit logs, OTP mismatch records, and wallet adjustments.
- **AI & Bot Operations:** Monitors Groq LLM token consumption and WhatsApp webhook performance metrics.

TECHNOLOGY STACK & FRAMEWORKS

Tools & Technologies Identified

A resilient, decoupled architecture leveraging modern Python web frameworks, polyglot persistence, and edge AI.

BACKEND FRAMEWORKS & APIS

- **Python Flask (Marketplace Core):** Handles application routing, Jinja2 template rendering, authentication sessions, and business logic.
- **FastAPI (Asynchronous AI Microservice):** High-throughput asynchronous ASGI microservice managing WhatsApp Cloud webhooks and AI model pipelines.
- **Uvicorn ASGI Engine:** Powers lightning-fast asynchronous event loops for AI diagnostic endpoints.

DATABASE INFRASTRUCTURE (HYBRID)

- **PostgreSQL (Relational Store):** ACID-compliant storage for structured marketplace entities: Users, Farms, Products, Orders, Wallets, and OTP records.
- **MongoDB (Document Store):** Flexible, schemaless JSON document store archiving multimodal chat transcripts, diagnostic images, and audio metadata.
- **Dual-DB Optimization:** Separates heavy transactional financial ledgers from unstructured conversation history.

REAL-TIME & SPATIAL CAPABILITIES

- **Flask-SocketIO:** Drives full-duplex bi-directional WebSocket communication for instantaneous order pings, farmer alerts, and driver coordinate streaming.
- **Leaflet.js & OpenStreetMap:** Client-side interactive vector maps rendering live GPS tracking, delivery radius buffers, and routing path lines.
- **Atomic Stock Management:** Prevents race conditions and duplicate order placement during high-traffic checkouts.

INTELLIGENT AI & SPEECH PROCESSING

- **Groq Cloud API (Qwen Model):** Sub-second LPU inference powering contextual crop pathology advisories, organic remedies, and prevention tips.
- **OpenAI Whisper (Speech-to-Text):** State-of-the-art neural STT transcribing spoken vernacular audio notes from WhatsApp.
- **Microsoft Edge-TTS:** High-fidelity neural speech synthesis delivering voice responses back to rural farmers.
- **PyTorch & Pillow:** Computer vision image normalization and deep learning leaf disease classification.

Transactional Lifecycle & AI Diagnostic Workflow

Linear step-by-step trace of the e-commerce transaction lifecycle and multimodal AI advisory loop.

WORKFLOW A: E-COMMERCE TRANSACTIONAL LIFECYCLE (FARMFETCH)

Step 1: Order Placement
Customer discovers farm, adds fresh produce to cart, and selects COD checkout with geo-coordinates.

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Step 2: Atomic Stock Lock
PostgreSQL transaction locks rows, atomically updates inventory, and prevents double-selling.

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Step 3: Real-Time Ping
Flask-SocketIO emits instant audio-visual order notification to Farmer's dashboard.

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Step 4: Driver Tracking
Nearby driver accepts order; Leaflet.js tracks transit; driver arrives and verifies Pickup OTP.

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Step 5: Dual OTP & Split
Customer validates Delivery OTP; automated wallet split executes (92% farmer, 85% driver fee).

WORKFLOW B: KRISHIMITRA AI DIAGNOSTIC & ADVISORY PIPELINE

Step 1: WhatsApp Query
Farmer uploads text, voice note, or diseased leaf photograph to KrishiMitra WhatsApp bot.

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Step 2: Media Processing
FastAPI webhook receives payload; Whisper converts speech to text; Pillow normalizes image matrix.

→

Step 3: Vision Pathology
Hugging Face / PyTorch vision model analyzes pathology patterns and classifies crop disease.

→

Step 4: Groq LLM Advisory
Qwen 3.8-27B model synthesizes organic remedies, chemical spray dosage, and prevention roadmap.

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Step 5: Voice & Multimodal
Edge-TTS synthesizes natural audio response; bimodal reply delivered to WhatsApp; logged to MongoDB.

Sustainability Focus: SDG & ESG Compliance

Quantifiable alignment with UN Sustainable Development Goals and Environmental, Social, and Governance criteria.

UN SDG 2: ZERO HUNGER

Target 2.3: Doubling Smallholder Incomes

- **Direct Target 2.3 Impact:** Explicitly supports the UN mandate to double agricultural productivity and incomes of small-scale food producers.
- **Eliminating Intermediary Extraction:** Redirects up to 30-40% of retail margins back to cultivating families, strengthening rural economic resilience.
- **Equal Technology Access:** Democratizes advanced AI agronomist tools to smallholders regardless of farm size or geographic isolation.
- **Crop Loss Mitigation:** Rapid pathology intervention protects baseline yields from disease-induced devastation.

ENVIRONMENTAL IMPACT (E)

Minimizing Carbon & Food Miles

- **Drastic Food Mile Reduction:** Localizing produce distribution within peri-urban clusters cuts long-haul diesel transport by over 60%.
- **Reduced Carbon Footprint:** Direct routing eliminates centralized transit hubs, multiple vehicle re-loadings, and wasted transit hours.
- **100% Paperless Operations:** Digital cataloguing, electronic OTP handshakes, and virtual wallet receipts eliminate paper manifests and billing waste.
- **Organic Remedy Promotion:** KrishiMitra AI prioritizes biological and organic crop treatments, curbing soil and groundwater chemical pollution.

SOCIAL IMPACT (S)

Digital & Linguistic Inclusion

- **Bridging the Rural Digital Divide:** Voice-first query interface empowers non-technical, semi-literate farmers who cannot type in conventional apps.
- **Multilingual Vernacular Support:** Whisper and Edge-TTS break English-only linguistic barriers, communicating in regional spoken tongues.
- **Frictionless WhatsApp Access:** Operates via a platform farmers already trust, removing complex app installation and login hurdles.
- **Fair Driver Livelihoods:** Transparent 85% delivery fee splits and immediate wallet payouts provide dignified local employment opportunities.

Challenges Faced, Limitations & Next Steps

Hurdles overcome during engineering, current operational boundaries, and planned evolutionary roadmap.

CHALLENGES OVERCOME

Engineering Solutions Implemented

- **High-Volume Race Conditions:** Resolved checkout race conditions during simultaneous orders via PostgreSQL atomic inventory decrement triggers and row-level locks.
- **Asynchronous Voice Streaming:** Engineered resilient audio streaming pipelines over constrained rural 3G/4G bandwidths using background threading and buffer caching.
- **Polyglot State Sync:** Synchronized relational transactional state in PostgreSQL with schemaless conversational transcripts in MongoDB without latency overhead.

CURRENT LIMITATIONS

Scope Boundaries of Prototype

- **Simulated GPS Navigation:** Delivery driver route progression currently operates via client-side coordinate interpolation within Leaflet browser scopes rather than native GPS hardware daemons.
- **Cash-on-Delivery (COD) Focus:** Platform currently executes financial wallet splits within an internal virtual ledger, pending live commercial banking integration.
- **Server-Side Inference Load:** Speech recognition (Whisper) and image preprocessing demand dedicated compute resources on the local host instance.

FUTURE ROADMAP

Next-Generation Enhancements

- **Payment Gateway Integration:** Integrating live Razorpay / UPI AutoSplit APIs to facilitate real-time NEFT/IMPS bank payouts directly into farmers' and drivers' savings accounts.
- **ML Crop Price Prediction:** Incorporating machine learning time-series forecasting (LSTM / Prophet) to advise farmers on optimal seasonal harvest pricing.
- **Native Driver Mobile App:** Porting logistics features into native Android/iOS mobile environments (Flutter/React Native) for continuous background hardware GPS pinging and geofencing.

THANK YOU!

Digital Platform for Direct Marketing of Local Farm Produce

An Integrated Agricultural Dual-Capability Platform: FarmFetch + KrishiMitra AI

💰 92% Direct Farmer Realization | ⚡ Sub-Second AI Agro-Diagnosis | 🌱 Zero-Waste Digital Supply Chain

QUESTIONS & TECHNICAL DISCUSSION

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